Impact of Locational Marginal Electricity Pricing on future Industrial Energy Systems in Germany

Felix Flatter
Department of Electrical Engineering
University of Kaiserslautern-Landau
Kaiserslautern, Germany
ORCID: 0009-0006-6286-4949

Lisa Reis
Department of Electrical Engineering
University of Kaiserslautern-Landau
Kaiserslautern, Germany

Stefan Goetz
Department of Electrical Engineering
University of Kaiserslautern-Landau
Kaiserslautern, Germany

***Abstract*—System and market operators use dynamic electricity prices to incentivize customers to adjust their active power exchange according to the current scarcity of electricity. Additionally, regulators and policymakers across Europe discuss market reforms introducing location-dependent electricity prices with varying granularity to reflect regional differences in electricity scarcity. As the 2022 energy price crisis in Europe highlighted, industrial profitability is highly sensitive to electricity prices. Therefore, industrial customers have reservations about regional electricity prices, especially those customers in potentially more expensive regions. In this paper, we explore the effects of dynamic and locational electricity prices on a hypothetical industry plant using locational marginal prices for the German electricity system for 2050 from the literature. Using a stylized model of the hypothetical industrial plant and its flexibility options as well as synthetic load profiles, we determine the optimal configuration of the internal energy system, energy flows, use of flexibility as well the the cost of energy for the plant. We find that leveraging electricity price spreads with energy flexibility can compensate for higher mean electricity prices and therefore alleviate locational disadvantages for industrial customers. Therefore, introducing locational electricity prices is not necessarily a disadvantage for customers—given an appropriate regulatory framework.**

***Index Terms*—locational marginal pricing, industrial energy systems, dynamic energy pricing**

I. Introduction

The increasing share of renewable energy generation, the introduction flexible technologies with additional degrees of freedom, and delayed grid enhancement make the future's power system more complex to plan and operate. Researchers, regulators, and policymakers advocate for unbiased electricity prices as a simple yet efficient means to manage this new complexity. Such unbiased electricity prices incentivize customers to both invest in own generation and flexible technologies as well as to operate their technologies optimally with regard to the power system, market, and grid. Two components are necessary to ensure the effects of unbiased electricity prices: the electricity price must reflect the changing value of electricity both over time—*dynamic pricing*—and in different areas—*regional or locational pricing* [1], [2].

Dynamic electricity pricing is already state-of-the-art around the world as spot markets have been operating on hourly or even shorter time bases for many years. Flexible processes and technologies allow customers to shift parts of their electricity demand from times of high to times of low prices, lowering operational costs. Large customers often already buy electricity directly from spot markets, and more and more smaller customers follow as intelligent energy management systems and flexible technologies become more attractive [1].

In Germany, the spot market electricity price is identical in the whole country as the zonal market only has a single bidding zone. Therefore, the electricity price does reflect neither the spatial diversity of renewable energy generation nor limited transmission capacities, causing grid operators to intervene in the market result with expensive congestion management methods to maintain operational limits in the power grid. Researchers, regulators, and policymakers discuss splitting the German bidding zone into two or more smaller bidding zones [3]. Each bidding zone then settles its own electricity price depending on the generation and load within the bidding zone, the cost of generation, and the available connections to other bidding zones. Researchers have also proposed a market based on *locational marginal pricing (LMP)* that assigns a dynamic electricity price to each node of the power grid [4], [5]. With LMP, each node is its own bidding zone, and the electricity price formation works in principle the same way as in the zonal market with several bidding zones. However, the LMP market result is inherently congestion free [6].

While dynamic electricity pricing has become state-of-the-art, many customers have reservations towards locational pricing due to potentially higher electricity price levels. In this paper, we set out to show that the combination of dynamic and locational electricity prices with flexible technologies and own generation can compensate for higher mean electricity price levels. We built a high-level optimization model of a hypothetical industrial plant considering its power and heat demand, own conventional and renewable generation, and flexibility. Then, we subjected this model to a constant reference electricity price as well as four extreme locational dynamic electricity prices from a literature LMP model for 2050 [6]. Finally, we compared the economically optimal configuration of the internal energy system as well as the optimal operation over the year 2050 for each case to infer recommendations for future investment decisions.

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TABLE I
ELECTRICITY PRICE CHARACTERISTICS

Scenario	Minimum €/MWh	Mean €/MWh	Maximum €/MWh	Max. Spread €/MWh
Minimum mean	1.00	162.23	362.15	361.15
Maximum mean	4.50	262.21	576.00	571.50
Minimum spread	123.01	219.41	382.08	259.07
Maximum spread	1.00	225.44	1581.51	1580.51

TABLE II
GAS PRICE COMPONENTS

Component	Value	Source
Energy cost	81.52 €/MWh	[11, part B, p. 224]
Non-deductible taxes & levies	14.42 €/MWh	[10, p. 105]
Grid fees	10 €/MWh	[11, part B,p. 260]
Total	105.95 €/MWh	

In Section II, we describe the methodology, the data used for the simulations, and the optimization model. Subsequently, we present the resulting installed capacities, energy flows, and energy costs for the selected use-cases in Section III. We draw conclusions about the implications for investments in industrial energy systems in Section IV, discuss limitations of our model and give an outlook on further research.

II. METHODOLOGY

A. Energy price time series

1) Electricity price: We selected four nodes from the LMP-based cellular energy market (CEM) proposed in [6]–[9] to represent the four extreme cases of locational and dynamic electricity pricing: the two nodes with the highest and lowest average price as well as the two nodes with the smallest and largest price spread. Table I lists the minimum, mean, and maximum electricity prices as well as the maximum spread for the selected nodes. The given electricity prices include grid fees as well as non-deductible taxes and levies as assumed in current literature Agency [10], [11]. However, the electricity prices do not include third-party service fees or energy subsidies.

For comparison, we also used the constant electricity price of 216 €/MWh including the same grid fees, taxes, and levies as the dynamic electricity prices given by a 2018 study by the German Energy Agency [11, part B, p. 254, 259].

2) Gas price: Although markets for gaseous energy carriers also include exchange-based trading with shorter trading intervals, we considered our hypothetical industry plant to procure gas or hydrogen at a fixed price for the entire year, e.g., through contracts-for-difference or an appropriate hedging strategy. Table II shows the price components for the gas mix in the public gas grid for 2050.

3) District heating price: The district heating market is usually controlled by a single entity that is also the grid operator, simplifying the price formation. We used the district heating price of 45 €/MWh from [11, part B, p. 382].

TABLE III
LOAD FACTORS, POWER, AND ANNUAL ENERGY DEMAND FOR TIME SERIES SYNTHESIS

	Normal	Weekends, holidays, shutdown periods	Nights	breaks	Peak power	Annual energy demand
Electricity	1	0.35	0.5	0.8	25 MW	140 GWh
Room heat	1	0.3	0.3	0.3	30 MW	103 GWh
LT process heat	1	0.05	0.1	0.65	13 MW	52 GWh
HT process heat	1	0	0	0	12 MW	45 GWh

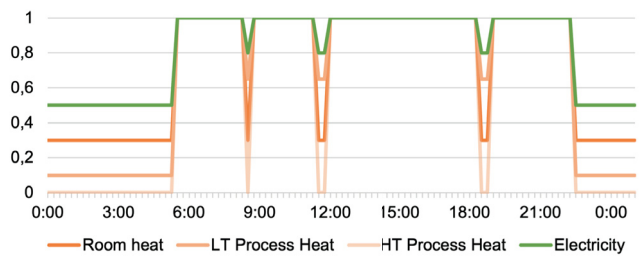

Fig. 1. Load profile for a weekday

B. Industrial load time series

The hypothetical industry plant has four loads: electrical, room heat at 60 °C, low-temperature (LT) process heat at 120 °C, and high-temperature (HT) process heat at 600 °C (see Fig. 3). The plant is shut down on weekends, on public holidays, before January 6th, 2050, and after December 22nd, 2050. Fig. 1 shows the regular weekday load profile. We synthesized normalized load profiles for all four load types on a 15-minute scale using the factors given in Table III. To account for seasonal influences, we modulated all time-series data except HT process heat with sinusoidal adjustment functions (see Fig. 2). Finally, we scaled the normalized load time series using the peak loads derived from three randomly selected environmental reports of German car manufacturers given in Table III [12]–[14].

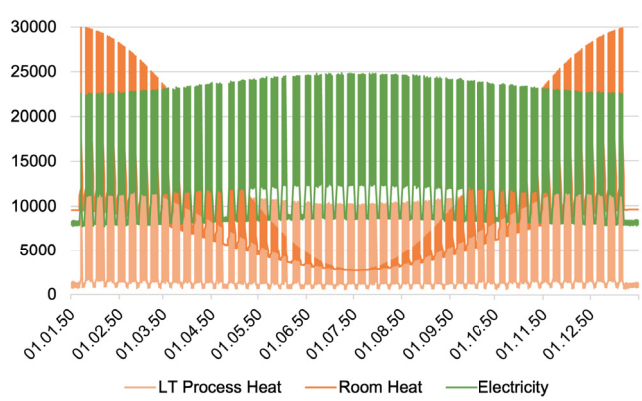

Fig. 2. Load profiles for 2050 (HT process heat omitted)

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TABLE IV
SUPPLY TECHNOLOGY OPTIONS

Energy demand type	Supply technology options
Electricity	Combined heat and power, photovoltaic, grid supply
Room heat	Combined heat and power, gas heating, heat pump, grid supply
LT process heat	Combined heat and power, heat pump, gas heating
HT process heat	Direct electric heating (Power-to-heat), direct gas-fired heating

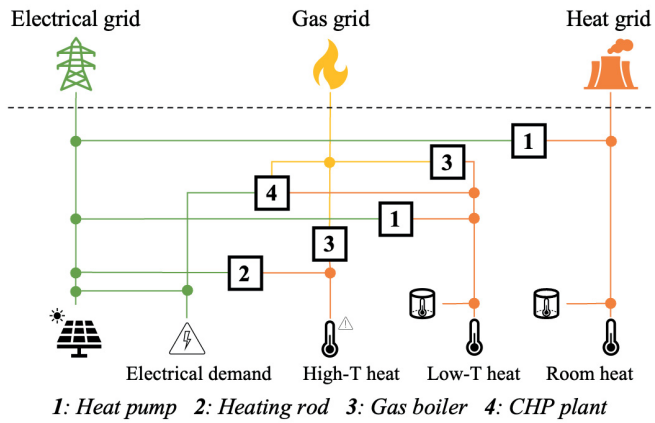

Fig. 3. Industry plant model

C. Technology models

The optimization model finds the optimal configuration of the internal energy system for the hypothetical industrial plant using the load time series presented in the previous section and a variety of different technology options to supply the load listed in Table IV. Other technologies are available, however, we selected only those with a sufficient level of maturity. Table V lists the relevant parameters for the selected technology options. Most parameters are literature values, but some missing values had to be estimated [11].

The heat pumps for both room heat and LT process heat use river water as their heat reservoir. We calculate the real coefficient of performance (COP) of both heat pump types by reducing the theoretical COP calculated from the flow and source temperatures by 40 % to account for practical limitations and losses:

$$COP_{\text{real}} = 0.6 \cdot \frac{T_{\text{flow}}}{T_{\text{flow}} - T_{\text{source}}}. \tag{1}$$

D. Optimization model

Together, the load time series, energy price time series, and technology models form the optimization model formulated in the open-source energy modeling framework *Calliope* [15]. Calliope provides and interpreter with predefined technology model templates to compile the mathematical optimization model using Pyomo, xarray, and Pandas. The objective function was to minimize the overall cost of energy across all energy carriers considering both investment and operational cost. There was no objective for self-sufficiency.

TABLE V
SUPPLY TECHNOLOGY PARAMETERS

Technology	Max. power	Efficiency
Electric grid supply	80 MW	100 %
Gas grid supply	80 MW	100 %
District heating grid supply	80 MW	100 %
Photovoltaic	10 MW	-
Combined heat and power (CHP)	-	30 % elec., 50 % therm.
Gas heating	-	93 %
Heat exchanger	-	90 %
Heat pump	-	COP curve
Direct electric heating	-	99 %
Direct gas-fired heating	-	90 %
	CAPEX	**OPEX**
Electric grid supply	62.50 €/kW	-
Gas grid supply	75 €/kW	-
District heating grid supply	250 €/kW	-
Photovoltaic	547 €/kW	17 €/kWp/a
Feed-in tariff	-	90 €/MWh
Combined heat and power (CHP)	1020 €/kW	41 €/kW/a
Gas heating	60 €/kW	6 €/kW/a
Heat exchanger	10 €/kW	5 %/a
Heat pump	700 €/kW	5 %/a
Direct electric heating	100 €/kW	1 €/kW/a
Direct gas-fired heating	100 €/kW	1 €/kW/a
Battery storage	350 €/kWh	10 €/kWh/a
Thermal storage (room heat)	2.50 €/kWh	5 %/a
Thermal storage (LT process heat)	20 €/kWh	5 %/a
	P/E rating	**Efficiency**
Battery storage	1	90 %
Thermal storage (room heat)	0.15 - 1	85 %
Thermal storage (LT process heat)	0.15 - 1	85 %

III. SIMULATION RESULTS

A. Installed capacities

1) Grid connections: First, we compare the rated power of the hypothetical industry plant's grid connections in Fig. 4[1]. In the case of constant electricity prices, the shares of electricity, gas, and district heating are similar, with the electrical grid connection at 31 MW. Closest to the constant price scenario is the minimum spread scenario, however, the gas electrical grid connection is now rated at 37 MW. Comparing the gas and district heating grid connections in the minimum mean and maximum mean scenarios indicates that high electricity prices lead to a shift towards using gas or district heating. The exact balance of electricity, gas, and district heating depends mainly on the absolute price levels of the energy carriers. In all dynamic electricity price scenarios the electrical grid connection is dominant, showing a positive correlation between the electricity price spread and the rating of the electrical grid connection. Such a correlation—best shown in the maximum spread scenario—is to be expected from a flexible industry plant as it uses as much electricity as possible during times of low prices. Note, however, that the rating of the grid connection alone is not an indicator for the energy drawn from each grid (see Section III-B).

[1]The installed capacity of PV is 10 MW in all scenarios.

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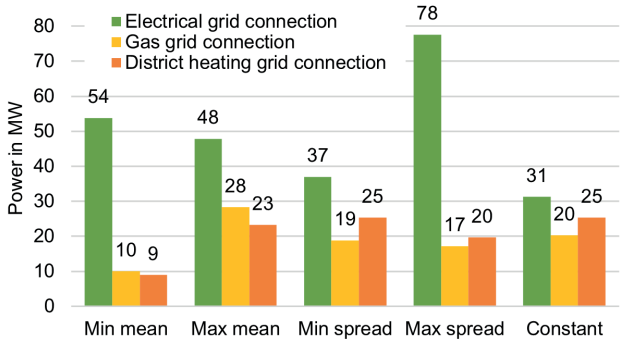

Fig. 4. Grid connection capacities by scenarios

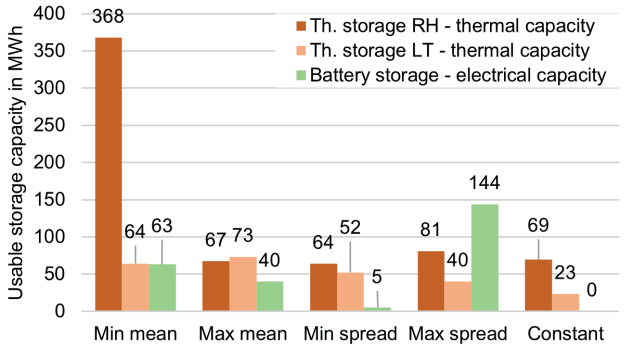

Fig. 5. Storage capacities by scenarios

2) Storage: Next, we compare the optimal storage capacity for each scenario in Fig. 5. Since there is no incentive to shift electrical or heat demand by means of storage in the constant price scenario, optimal storage capacities for this case are overall small compared to the dynamic price scenarios. In general, thermal storage for room heating increases the ability of the industrial plant to use electricity to generate heat during times of low prices through a heat pump and use this stored energy during times of high prices, even across longer periods of time. The ability to use thermal storage for room heating is determined by the absolute price level of electricity, as for example the storage capacity in the minimum mean scenario shows. On the other hand, thermal storage for LT process heat varies only slightly across all scenarios since the LT process heat demand is periodic and regular throughout the year. Remaining differences in LT process heat storage capacity across the scenarios are influenced slightly by mean electricity price levels as well as the coupling of LT process heat and room heating through the heat exchanger. For battery storage, comparing the minimum and maximum spread scenarios reveals a strong correlation between electricity price spread and battery capacity. Here, the added flexibility of battery storage enables the industrial plant to shift its electrical demand between times of low and high prices. Absolute price levels also have an influence on battery sizing, albeit the correlation is negative in this case.

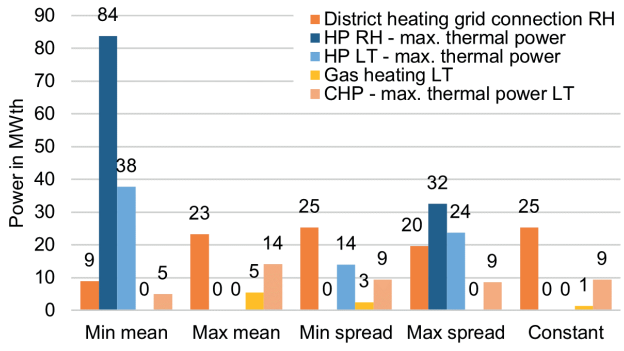

Fig. 6. Heat sources by scenario

3) Heat sources: Last, we compare the heat sources for room heating and LT process heat in Fig. 6. For heat pumps, we calculated the maximum thermal power from the installed electrical power and the maximum COP. Therefore, the maximum thermal power is not fixed but varies with the heat source temperature. The optimization model only deploys heat pumps when electricity is very cheap (minimum mean scenario), or a price difference can be exploited by means of thermal storage (maximum spread scenario). In all other cases, primary sources for room heating and LT process heat are district heating and a CHP plant, respectively. Comparing the constant price and minimum spread scenarios, we recognise the effect of even low price spreads on LT process heat generation—despite similar CHP and gas heating ratings, in the minimum mean scenario we see an additional heat pump and increased storage. We can also identify a negative correlation between direct gas-fired heating for LT process heat and mean electricity prices. HT process heat demand is served by direct electric heating in all scenarios, independent of electricity price changes.

B. Energy consumption and cost

1) Consumption by energy carrier: While analyzing the installed capacities of the energy supply technologies allows us to draw conclusions about the maximum power drawn from each grid and the associated operational strategies for storage and conversion technologies, comparing the energy consumption from each grid shows directly the balance between energy carriers within the industrial plant. We clearly identify the expected strong negative correlation between mean electricity prices and consumption in the minimum and maximum mean scenario in Fig. 7. Gas and district heating consumption behave inversely, mainly due to the use of heat pumps for room heating and LT process heat. Despite being the extreme cases for electricity consumption, the rated power of the grid connection is similar in the minimum and maximum mean scenario, indicating a limited dynamic of the electricity consumption. In comparison, the minimum and maximum spread scenarios show a very much different behavior. Here, we identify a positive correlation between mean electricity prices and consumption, contradicting the expectation formed in the previous comparison.

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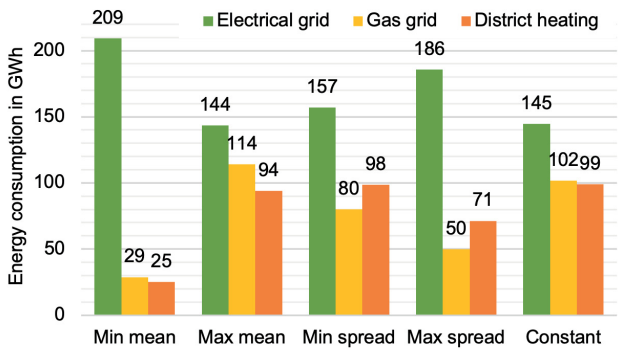

Fig. 7. Annual energy consumption from public grids by scenario

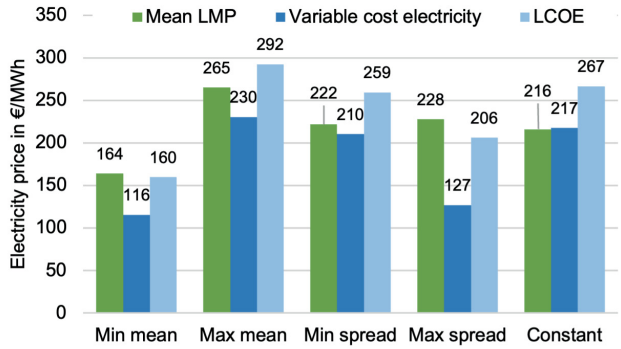

Fig. 8. Electricity prices by scenario

The explanation for this positive correlation lies in the flexibility provided by both the battery as well as the thermal storage in combination with heat pumps. This flexibility enables the industrial plant to take advantage of low electricity prices. In contrast to the previous comparison, the grid connection rating of the minimum and maximum spread varies by more than 200 % while the electrical energy consumption is similar, indicating a more dynamic electricity consumption behavior. Furthermore, comparing the constant price and maximum mean scenarios shows that using flexibility can make electricity more attractive over other energy forms as the electricity consumption is almost identical despite a 46 € difference in mean electricity prices.

2) Variable and levelized costs: Finally—and perhaps most importantly—we consider the cost of energy for our industrial plant, given in Fig. 8. We compare the mean electricity price, the actual variable cost for electricity from the grid, and the levelized cost of electricity (LCOE) also considering the annualized investments. Variable costs for electricity are lower in all dynamic price scenarios compared to the mean electricity price, confirming the hypothesis that dynamic prices and flexible demand can lower electricity cost—a relation particularly visible in the minimum and maximum spread scenarios. Leveraging price dynamics with flexibility can even push the LCOE below the mean electricity price despite including the annualized cost of the necessary electrical storage infrastructure. However, for disadvantageous locations with high mean prices or limited price dynamics, the benefits are also limited, for example in the maximum mean and minimum spread scenarios.

IV. Discussion

Our main finding is that industry plants can leverage dynamic electricity prices through flexibility to achieve overall lower energy cost and to compensate for higher electricity prices in disadvantageous locations. We showed that in some cases deployment of flexibility can push the plant's LCOE below the mean electricity price despite significant investment cost for storage. Such a procurement strategy adequate to dynamic and locational electricity pricing makes electricity more attractive as an energy carrier, even when mean electricity prices are higher than in the reference scenario. Therefore, investments into flexible technologies and processes can be the key to long-term profitability of both existing and future industry plants, especially if electricity prices become dependent on locations in the future.

For policymakers, the results of our work imply two considerations for potential decisions on locational electricity pricing. First, introducing locational or regional electricity pricing does not necessarily imply higher electricity cost for customers in a by mean price disadvantageous area—if these customers invest in flexibility. Second, the regulatory framework for such investments in flexibility must match the steps towards locational electricity prices so that customers can leverage their flexibility as early as possible. This regulatory framework must specifically include a long-term strategy and deployment schedule for all measures.

We identified several key indicators relevant for industrial investment decisions from the simulations:

1) In general, energy price levels influence the attractiveness of different energy carriers over each other. Therefore, stakeholders must take the expected mean price levels for electricity, gas/hydrogen, district heating, and other supply options into account when weighing supply technology options or choosing new locations.
2) However, the use of flexibility can push the actual electricity cost below the mean electricity price.
3) The optimal electrical grid connection rating correlates positively with the electricity price spread while the grid connection rating of other energy carriers correlates negatively. However, the energy consumed from the electricity grid correlates positively to the mean electricity price and *vice versa* for gas and district heating. We can conclude that low mean electricity prices lead to a lower, but more constant electricity demand and that a high electricity price spread leads to a more dynamic electricity demand, potentially resulting in additional congestion management challenges for the grid operator. Grid operators should thus be included early in planning processes to ensure the feasibility of the desired grid connection rating.

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4) Battery storage sizing depends mainly on the electricity price spread with mean electricity prices as a secondary influence. Therefore, stakeholders should estimate electricity price dynamics, e.g., by obtaining information about regional renewable generation, grid congestions, and ramp-up of new electrical loads.
5) There are two main contributors to thermal storage sizing. First, higher thermal load dynamics imply larger storage to equalize generation from for example heat pumps. It is therefore imperative for stakeholders to consider not only the overall thermal energy demand, but the time characteristics of (future) processes. Second, seasonally low electricity prices incentivize long-term storage, especially when using heat pumps. Here, forecasts of regional renewable energy generation provide the necessary information to stakeholders, e.g., in regions dominated by PV generation.

In conclusion, planning future industrial energy systems require the responsible parties to not only consider mean energy prices and internal optimizations, but also the expected dynamics and regional characteristics of electricity prices. Although policymakers have not yet decided for or against introducing location-dependent electricity pricing, taking regional renewable energy generation, load projections, and grid planning into account when planning new industrial developments can provide a basis at least mitigate challenges of potential regulatory changes to electricity pricing. As a general rule, the more detailed and extensive the available information on regional, national, and European changes in the energy system are, the better planning decisions can be.

One aspect of flexibility we did not yet consider in our model is the actual flexibility of production processes themselves as we did not have a sufficient data source for building such a model. However, given the conclusions drawn from our simulations, we expect the effects of adding production process flexibility to be similar to the effects of battery and thermal storage—albeit, at potentially much lower investment and operational cost. Since identifying and quantifying process flexibility is too complex and intricate to be considered here, we refer the reader to the recent works of Fraunhofer IPA on this topic [16], [17].

Another limitation in our methodology is the use of a linear cost model for technology ratings when in reality the total investment cost for a technology is comprised of a constant part (e.g., planning and property cost) and a variable part proportional to the power rating. However, our model is not intended to provide accurate cost calculations for investment decisions, but rather to compare different energy pricing scenarios. Additionally, the exact ratio of the constant and variable parts of the investment cost greatly depend on individual circumstances and vary from case to case.

In the future, we plan on coupling the optimization model presented here with anonymized industrial standard load profiles in a seamless industrial load time series tool chain for stakeholders and researchers. The tool chain is going to allow users to synthesize industrial load profiles using either pre-processed standard load profiles or individual raw data and to then explore different supply options and energy cost scenarios for their individual use-case.

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